



Holmes CTF 2026: The Reichenbach Directive - AfterParty

Holmes CTF 2026 - Paper Ghost

Category: DFIR / Forensics

Event: Holmes CTF 2026 - The Reichenbach Directive (HackTheBox)

Scenario: 4 - Paper Ghost

Tools used: python-registry, RegRipper, custom Python script



Setup

Downloaded PaperGhost.zip, extracted it:

```
mkdir -p ~/PaperGhost
```

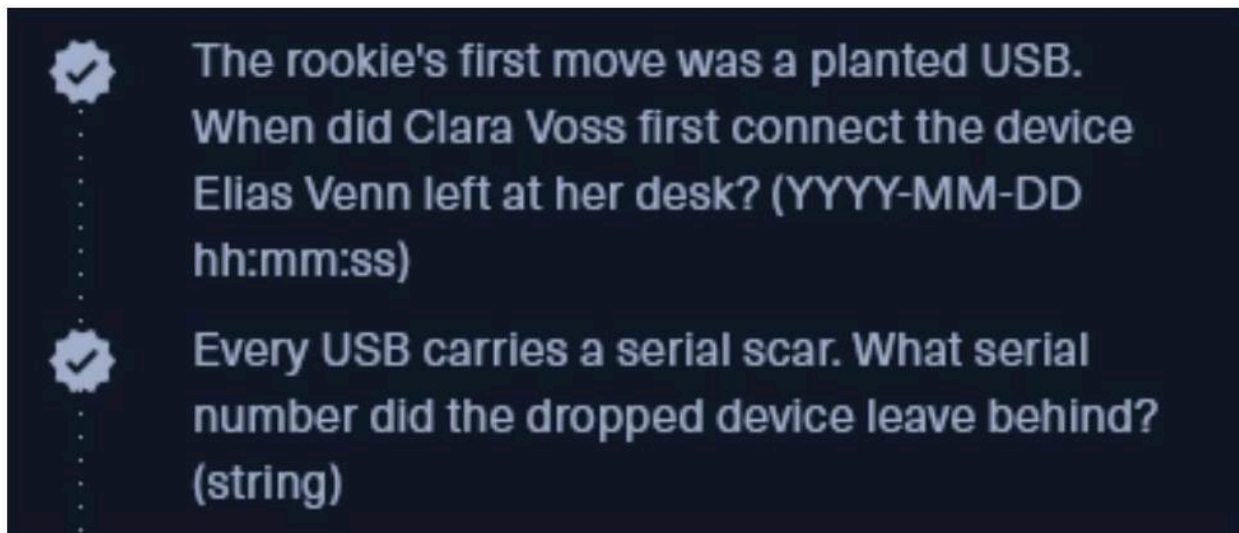
```
unzip ~/Downloads/PaperGhost.zip -d ~/PaperGhost
```

Inside: a PDF (case briefing) + a Triage folder - a forensic disk image extract containing:

- C:registry hive
- Users.DAT (user registry hive for Clara Voss)
- Console logs, skip logs, copy logs

Flag 1 & 2 - USB Connection Time + Serial Number

Question: When did Clara Voss first connect the USB, and what was its serial number?



Approach

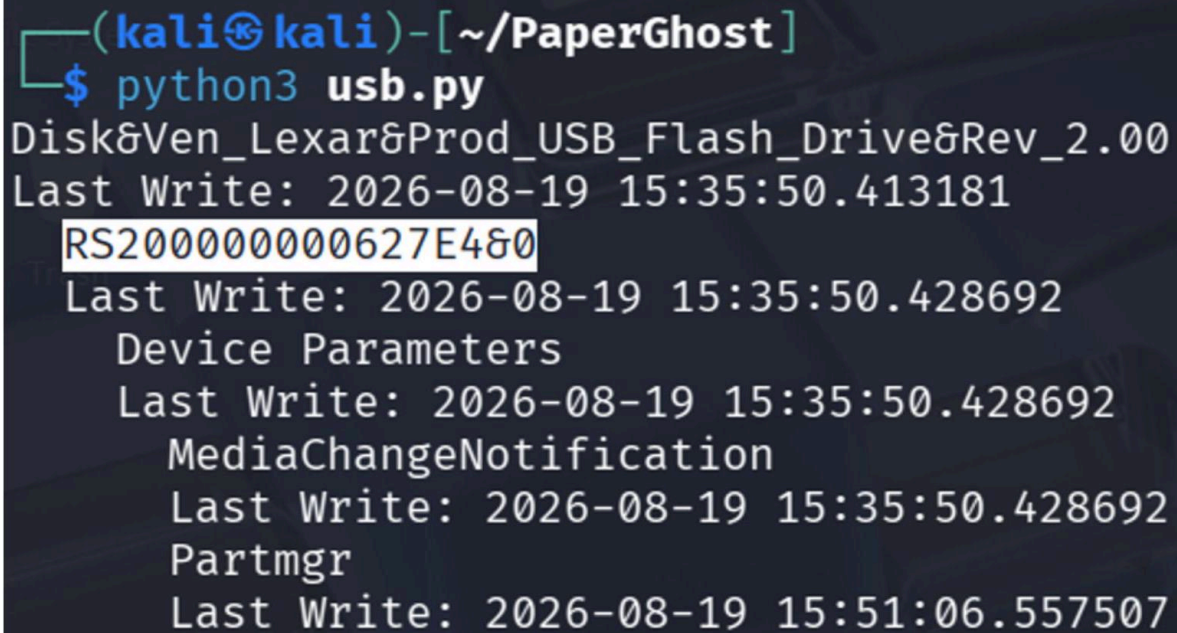
- The SYSTEM hive stores USB connection history under ControlSet001.
- Wrote a small Python script (usb.py) using the python-registry library to open the hive and recursively walk the USBSTOR subkeys, printing each key name and its last-write timestamp:

```
from Registry import Registry
hive = Registry.Registry("./PaperGhost/Triage/C/Windows/System32/config/SYSTEM")
key = hive.open(r"ControlSet001\Enum\USBSTOR")
def show_keys(key, indent=0):
```

```

    for subkey in key.subkeys():
        print(" " * indent + subkey.name())
        print(" " * indent + "Last Write:", subkey.timestamp())
        show_keys(subkey, indent + 2)
show_keys(key)
Ran it:
python3 usb.py
Output showed the device:
Disk&Ven_Lexar&Prod_USB_Flash_Drive&Rev_2.00
Last Write: 2026-08-19 15:35:50.413181
    RS200000000627E4&0
Last Write: 2026-08-19 15:35:50.428692

```



```

(kali@kali)-[~/PaperGhost]
$ python3 usb.py
Disk&Ven_Lexar&Prod_USB_Flash_Drive&Rev_2.00
Last Write: 2026-08-19 15:35:50.413181
    RS200000000627E4&0
    Last Write: 2026-08-19 15:35:50.428692
    Device Parameters
    Last Write: 2026-08-19 15:35:50.428692
    MediaChangeNotification
    Last Write: 2026-08-19 15:35:50.428692
    Partmgr
    Last Write: 2026-08-19 15:51:06.557507

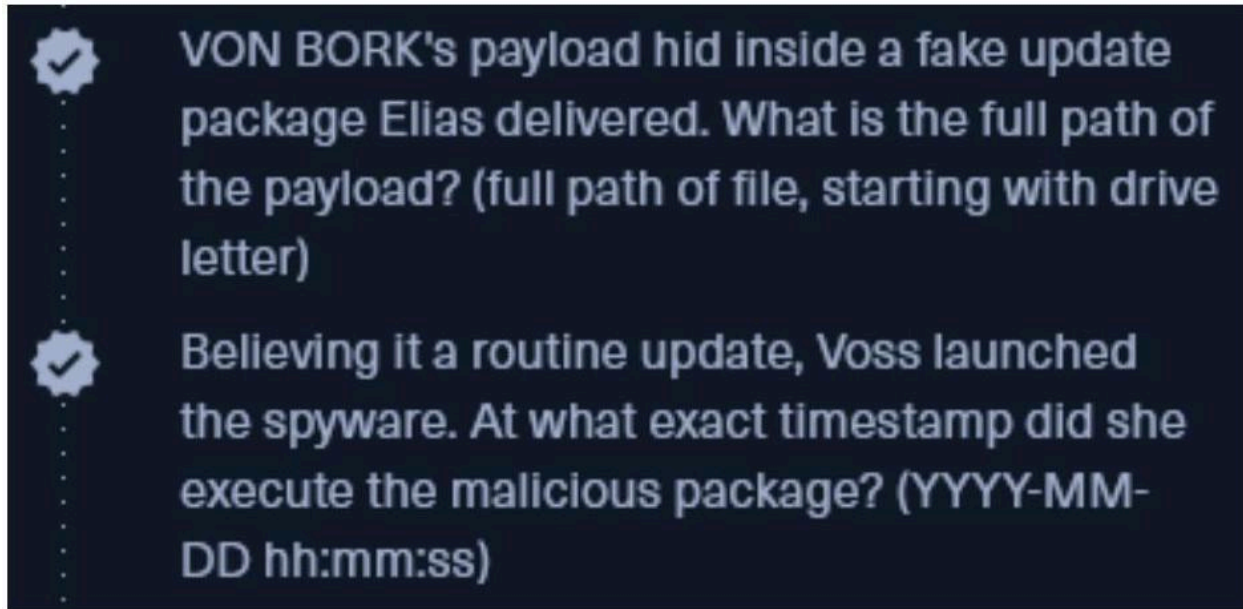
```

Answers found

- First connect timestamp (Flag 1): 2026-08-19 15:35:50 (first-write time of the device key)
- Serial number (Flag 2): RS200000000627E4 (from the instance ID subkey, &0 suffix trimmed since it just marks a non-unique serial)

Flag 3 & 4 - Malicious Payload Path + Execution Time

Question: What was the full path of the fake update payload, and when did Voss execute it?



Approach

- Ran a change of directory into the triage folder and used RegRipper to parse NTUSER.DAT for user cvoss with the userassist plugin - this tracks GUI-launched programs with execution counts and last-run times:

```
cd ~/PaperGhost/PaperGhost/Triage/C
sudo perl /usr/lib/regripper/rip.pl \
-r Users/cvoss/NTUSER.DAT \
-p userassist
```

- Scrolled through the UserAssist output (program execution history) and found a suspicious entry launched from a USB drive letter (E:):

2026-08-19 15:36:25Z

E:\CO-LT-0469 update package update.exe (1)

This lined up perfectly with the timeline - it ran shortly after the USB was plugged in (15:35:50) and disguised itself as an "update package."

```
(kalis/kali)-[~/PaperGhost/PaperGhost/Triage/C]
$ sudo perl /usr/lib/regripper/rip.pl \
-r Users/cvoss/NTUSER.DAT \
-p userassist
[sudo] password for kali:
Launching userassist v.20170204
UserAssist
```

```
Software\Microsoft\Windows\CurrentVersion\Explorer\UserAssist
LastWrite Time 2026-08-19 12:49:08Z
{9E04CAB2-CC14-11DF-BB8C-A2F1DED72085}
{A3D53349-6E61-4557-8FC7-0028EDCEEBF6}
{B267E3AD-A825-4A09-82B9-EEC22AA3B847}
{BCB48336-4DDD-48FF-BB0B-D3190DACB3E2}
{CAA59E3C-4792-41A5-9909-6A6A8D32490E}
{CEBFF5CD-ACE2-4F4F-9178-9926F41749EA}
2026-08-19 15:50:46Z
    Microsoft.Windows.Explorer (6)
2026-08-19 15:38:27Z
    MSEdge (6)
2026-08-19 15:36:25Z
    E:\CO-LT-0469 update package\update.exe (1)
2026-08-19 13:41:24Z
    {1AC14E77-02E7-4E5D-B744-2EB1AE5198B7}\cmd.exe (2)
```

Answers found

- Payload full path (Flag 3): E:\CO-LT-0469 update package update.exe
- Execution timestamp (Flag 4): 2026-08-19 15:36:25